

Geography

This section presents the requirements for programs in:

- **Geography B.A. Honours**
- **Geography with Concentration in Physical Geography B.A. Honours**
- **Geography with Concentration in Urban Geography B.A. Honours**
- **Geography B.A. Combined Honours**
- **Geography B.A.**
- **Earth Sciences and Physical Geography B.Sc. Combined Honours**
- **Physical Geography B.Sc. Honours**
- **Specialization in Globalization and the Environment B.G.In.S. Honours**
- **Stream in Globalization and the Environment B.G.In.S.**
- **Minor in Geography**
- **Minor in Physical Geography**
- **Minor in Urban Studies**

Program Requirements

Geography

B.A. Honours (20.0 credits)

A. Credits Included in the Major CGPA (10.0 credits)

1. 1.5 credits in:	1.5
GEOG 1010 [0.5]	Global Environmental Systems
GEOG 1020 [0.5]	People, Places and Environments
GEOM 1004 [0.5]	Maps, Satellites and the Geospatial Revolution
2. 0.5 credit from:	0.5
GEOG 2020 [0.5]	Ecosystems of Canada
GEOG 2013 [0.5]	Weather and Water
GEOG 2014 [0.5]	The Earth's Surface
3. 1.0 credit in:	1.0
GEOG 2005 [0.5]	Introduction to Qualitative Research
GEOG 2006 [0.5]	Introduction to Quantitative Research
4. 1.5 credits from:	1.5
GEOG 2023 [0.5]	Cities, Inequality and Urban Change
GEOG 2200 [0.5]	Global Connections
GEOG 2300 [0.5]	Space, Place and Culture
GEOG 2500 [0.5]	Climate Change: Social Science Perspectives
5. 0.5 credit from:	0.5
GEOG 3000 [0.5]	Honours Field Course
GEOG 3030 [0.5]	Regional Field Excursion
6. 0.5 credit from:	0.5
GEOG 3001 [0.5]	Doing Qualitative Research
GEOG 3003 [0.5]	Quantitative Geography
GEOM 2007 [0.5]	Vector GIS: Points, Lines and Polygons
GEOM 3002 [0.5]	Introduction to Remote Sensing
GEOM 3007 [0.5]	Cartographic Theory and Design

7. 1.0 credit from:	1.0
GEOG 3021 [0.5]	Geographies of Culture and Identity
GEOG 3022 [0.5]	Environmental and Natural Resources
GEOG 3023 [0.5]	Cities in a Global World
GEOG 3024 [0.5]	Understanding Globalization
GEOG 3025 [0.5]	Geographies of Selected Regions
GEOG 3026 [0.5]	Topics in the Geography of Canada
GEOG 3206 [0.5]	Health, Environment, and Society
GEOG 3209 [0.5]	Sustainability and Environment in the South
GEOG 3501 [0.5]	Geographies of the Canadian North
8. 1.0 credit in GEOG and/or GEOM at the 3000- level or above	1.0
9. 2.5 credits from:	2.5
a) Thesis pathway:	
GEOG 4909 [1.0] plus 1.5 credits from GEOG/GEOM and/or ENST at the 4000-level	
b) Course pathway:	
2.5 credits from GEOG/GEOM and/or ENST at the 4000-level	
B. Credits Not Included in the Major CGPA (10.0 credits)	
10. 8.0 credits in electives not in GEOG	8.0
11. 2.0 credits in free electives	2.0
Total Credits	20.0

Geography with Concentration in Physical Geography

B.A. Honours (20.0 credits)

A. Credits Included in the Major CGPA (9.5 credits)

1. 1.5 credits in:	1.5
GEOG 1010 [0.5]	Global Environmental Systems
GEOG 1020 [0.5]	People, Places and Environments
GEOM 1004 [0.5]	Maps, Satellites and the Geospatial Revolution
2. 2.0 credits in:	2.0
GEOG 2005 [0.5]	Introduction to Qualitative Research
GEOG 2006 [0.5]	Introduction to Quantitative Research
GEOG 2013 [0.5]	Weather and Water
GEOG 2014 [0.5]	The Earth's Surface
3. 1.0 credit from:	1.0
GEOG 2023 [0.5]	Cities, Inequality and Urban Change
GEOG 2200 [0.5]	Global Connections
GEOG 2300 [0.5]	Space, Place and Culture
GEOG 2500 [0.5]	Climate Change: Social Science Perspectives
4. 0.5 credit in:	0.5
GEOG 3000 [0.5]	Honours Field Course
GEOG 3010 [0.5]	Field Methods in Physical Geography
5. 2.0 credits from:	2.0
GEOG 3003 [0.5]	Quantitative Geography
GEOG 3102 [0.5]	Geomorphology
GEOG 3103 [0.5]	Watershed Hydrology
GEOG 3104 [0.5]	Principles of Biogeography

GEOG 3105 [0.5]	Climate and Atmospheric Change	
GEOG 3106 [0.5]	Aquatic Science and Management	
GEOG 3108 [0.5]	Soil Properties	
GEOM 3002 [0.5]	Introduction to Remote Sensing	
6. 2.5 credits from:		2.5
a) Thesis pathway:		
i. 1.0 credit in:		
GEOG 4909 [1.0]	Honours Research Thesis	
ii. 1.5 credits from:		
GEOM 4003 [0.5]	Remote Sensing of the Environment	
GEOG 4004 [0.5]	Environmental Impact Assessment	
GEOG 4005 [0.5]	Directed Studies in Geography	
GEOG 4013 [0.5]	Cold Region Hydrology	
GEOG 4017 [0.5]	Global Biogeochemical Cycles	
GEOG 4101 [0.5]	Two Million Years of Environmental Change	
GEOG 4103 [0.5]	Water Resources Engineering	
GEOG 4104 [0.5]	Microclimatology	
GEOG 4108 [0.5]	Permafrost	
OR		
b) Course pathway:		
2.5 credits from:		
GEOM 4003 [0.5]	Remote Sensing of the Environment	
GEOG 4004 [0.5]	Environmental Impact Assessment	
GEOG 4005 [0.5]	Directed Studies in Geography	
GEOG 4013 [0.5]	Cold Region Hydrology	
GEOG 4017 [0.5]	Global Biogeochemical Cycles	
GEOG 4101 [0.5]	Two Million Years of Environmental Change	
GEOG 4103 [0.5]	Water Resources Engineering	
GEOG 4104 [0.5]	Microclimatology	
GEOG 4108 [0.5]	Permafrost	
GEOG 4408 [0.5]	Practicum II	
B. Credits Not Included in the Major CGPA (10.5 credits)		
7. 8.0 credits in	electives not in GEOG	8.0
8. 2.5 credits in	free electives.	2.5
Total Credits		20.0

Geography with Concentration in Urban Geography

B.A. Honours (20.0 credits)

A. Credits included in the Major CGPA (11.0 credits)

1. 1.0 credit in:		1.0
GEOG 1010 [0.5]	Global Environmental Systems	
GEOG 1023 [0.5]	Introduction to Cities and Urbanization	
2. 0.5 credit from:		0.5
GEOM 1004 [0.5]	Maps, Satellites and the Geospatial Revolution	
GEOG 1020 [0.5]	People, Places and Environments	
3. 0.5 credit from:		0.5
GEOG 2013 [0.5]	Weather and Water	
GEOG 2014 [0.5]	The Earth's Surface	
GEOG 2020 [0.5]	Ecosystems of Canada	
4. 1.0 credit in:		1.0

GEOG 2005 [0.5]	Introduction to Qualitative Research	
GEOG 2006 [0.5]	Introduction to Quantitative Research	
5. 0.5 credit in:		0.5
GEOG 2023 [0.5]	Cities, Inequality and Urban Change	
6. 1.0 credit from:		1.0
GEOG 2200 [0.5]	Global Connections	
GEOG 2300 [0.5]	Space, Place and Culture	
GEOG 2500 [0.5]	Climate Change: Social Science Perspectives	
7. 0.5 credit from:		0.5
GEOG 3000 [0.5]	Honours Field Course	
GEOG 3030 [0.5]	Regional Field Excursion	
8. 0.5 credit from:		0.5
GEOM 2007 [0.5]	Vector GIS: Points, Lines and Polygons	
GEOG 3001 [0.5]	Doing Qualitative Research	
GEOM 3002 [0.5]	Introduction to Remote Sensing	
GEOG 3003 [0.5]	Quantitative Geography	
GEOM 3007 [0.5]	Cartographic Theory and Design	
9. 0.5 credit in:		0.5
GEOG 3023 [0.5]	Cities in a Global World	
10. 0.5 credit from:		0.5
GEOG 3021 [0.5]	Geographies of Culture and Identity	
GEOG 3022 [0.5]	Environmental and Natural Resources	
GEOG 3024 [0.5]	Understanding Globalization	
GEOG 3025 [0.5]	Geographies of Selected Regions	
GEOG 3026 [0.5]	Topics in the Geography of Canada	
GEOG 3206 [0.5]	Health, Environment, and Society	
GEOG 3209 [0.5]	Sustainability and Environment in the South	
GEOG 3501 [0.5]	Geographies of the Canadian North	
11. 1.0 credit in:		1.0
GEOG 4023 [0.5]	Seminar in Special Topics on the City	
GEOG 4323 [0.5]	Urban and Regional Planning	
12. 0.5 credit from:		0.5
AFRI 3004 [0.5]	The African City	
ARCU 3100 [0.5]	The Morphology of the City	
HRSJ 3002 [0.5]	Right to the City	
13. 0.5 credit from:		0.5
ARCH 4201 [0.5]	History of Modern Housing	
ARCU 4103 [0.5]	Cities	
ARCU 4300 [0.5]	Theories of Urbanism	
ARCU 4700 [0.5]	Urban Utopias	
ARCU 4801 [0.5]	Topics in Urbanism	
GEOG 4000 [0.5]	Field Studies (when offered with an urban theme)	
GEOG 4005 [0.5]	Directed Studies in Geography (with urban theme)	
GEOG 4007 [0.5]	Special Topics in Geography and Environmental Studies	
INDG 4001 [0.5]	Indigenous Urbanisms	
14. 2.5 credits from:		2.5
a) Thesis pathway:		

GEOG 4909 [1.0]	Honours Research Thesis	
1.5 credits in GEOG/GEOM and/or ENST at the 4000 level		
b) Course pathway:		
2.5 credits in GEOG/GEOM and/or ENST at the 4000 level		
B. Credits Not Included in the Major CGPA (9.0 credits)		
15. 7.5 credits in	electives not in GEOG	7.5
16. 1.5 credits in	free electives	1.5
Total Credits		20.0

Geography

B.A. Combined Honours (20.0 credits)

A. Credits Included in the Geography Major CGPA (7.0 credits)

1. 1.0 credit in:		1.0
GEOG 1010 [0.5]	Global Environmental Systems	
GEOG 1020 [0.5]	People, Places and Environments	
2. 0.5 credit from:		0.5
GEOG 2020 [0.5]	Ecosystems of Canada	
GEOG 2013 [0.5]	Weather and Water	
GEOG 2014 [0.5]	The Earth's Surface	
3. 1.0 credit from:		1.0
GEOG 2023 [0.5]	Cities, Inequality and Urban Change	
GEOG 2200 [0.5]	Global Connections	
GEOG 2300 [0.5]	Space, Place and Culture	
GEOG 2500 [0.5]	Climate Change: Social Science Perspectives	
4. 1.0 credit from:		1.0
GEOM 1004 [0.5]	Maps, Satellites and the Geospatial Revolution	
GEOG 2005 [0.5]	Introduction to Qualitative Research	
GEOG 2006 [0.5]	Introduction to Quantitative Research	
5. 1.5 credits in	GEOG and/or GEOM at the 3000- level or above	1.5
6. 1.0 credit in	GEOG and/or GEOM at the 4000- level	1.0
7. 1.0 credit in:		1.0

a) Thesis pathway

GEOG 4909 [1.0]	Honours Research Thesis	
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OR

b) Course pathway

1.0 credit in GEOG at the 4000-level

B. Additional Requirements (13.0 credits)		13.0
8. The requirements of the other Honours discipline must be satisfied		
9. Sufficient free electives to total 20.0 credits for the program.		
Total Credits		20.0

Geography

B.A. (15.0 credits)

A. Credits Included in the Major CGPA (7.0 credits)

1. 1.0 credit in:		1.0
GEOG 1010 [0.5]	Global Environmental Systems	
GEOG 1020 [0.5]	People, Places and Environments	
2. 0.5 credit from:		0.5

GEOG 2020 [0.5]	Ecosystems of Canada	
GEOG 2013 [0.5]	Weather and Water	
GEOG 2014 [0.5]	The Earth's Surface	
3. 1.0 credit from:		1.0
GEOM 1004 [0.5]	Maps, Satellites and the Geospatial Revolution	
GEOG 2005 [0.5]	Introduction to Qualitative Research	
GEOG 2006 [0.5]	Introduction to Quantitative Research	
4. 1.0 credit from:		1.0
GEOG 2023 [0.5]	Cities, Inequality and Urban Change	
GEOG 2200 [0.5]	Global Connections	
GEOG 2300 [0.5]	Space, Place and Culture	
GEOG 2500 [0.5]	Climate Change: Social Science Perspectives	
5. 1.0 credit in	GEOG and/or GEOM at the 2000- level or above	1.0
6. 2.5 credits in	GEOG and/or GEOM at the 3000- level or above	2.5
B. Credits Not Included in the Geography Major CGPA (8.0 credits)		
7. 6.0 credits in	electives not in GEOG	6.0
8. 2.0 credit in	free electives.	2.0
Total Credits		15.0

Course Categories for B.Sc. Geography

Lists of courses for all other categories (Science Continuation, Approved Experimental Science, Science Faculty Electives and Approved Arts or Social Sciences Electives) are located at the Academic Regulations for the B.Sc. page.

Earth Sciences and Physical Geography

B.Sc. Combined Honours (20.0 credits)

A. Credits Included in the Major CGPA (13.0 credits)

1. 1.0 credit in:		1.0
ERTH 1002 [0.5]	The Earth and Life Odyssey: A Journey Through Billions of Years	
GEOG 1010 [0.5]	Global Environmental Systems	
2. 1.0 credit in:		1.0
GEOG 2013 [0.5]	Weather and Water	
GEOG 2014 [0.5]	The Earth's Surface	
3. 2.5 credits in:		2.5
ERTH 2102 [0.5]	Mineralogy to Petrology	
ERTH 2106 [0.5]	Geochemistry	
ERTH 2312 [0.5]	Paleontology	
ERTH 2314 [0.5]	Sedimentation and Stratigraphy	
ERTH 2407 [0.5]	Structural Geology	
4. 0.5 credit in:		0.5
ERTH 2802 [0.5]	Field Geology I	
5. 1.0 credits in:		1.0
ERTH 3004 [0.5]	Igneous Petrology	
ERTH 3405 [0.5]	Geophysical Methods	
6. 0.5 credit from:		0.5
ERTH 3205 [0.5]	Physical Hydrogeology	
GEOG 3103 [0.5]	Watershed Hydrology	
7. 1.0 credit in:		1.0

ERTH 2004 [0.5]	Maps, Satellites and the Geospatial Revolution	
GEOM 3002 [0.5]	Introduction to Remote Sensing	
8. 2.0 credits from:		2.0
GEOG 3003 [0.5]	Quantitative Geography	
GEOG 3010 [0.5]	Field Methods in Physical Geography	
GEOG 3102 [0.5]	Geomorphology	
GEOG 3104 [0.5]	Principles of Biogeography	
GEOG 3105 [0.5]	Climate and Atmospheric Change	
GEOG 3106 [0.5]	Aquatic Science and Management	
GEOG 3108 [0.5]	Soil Properties	
9. 0.5 credit in:		0.5
ERTH 4302 [0.5]	Frozen Earth: Unveiling the Snowball Earth Catastrophe	
10. 1.0 credit in	Science Geography or Geomatics courses at the 2000-level or above	1.0
11. 1.0 credit in	Earth Sciences, Science Geography or Geomatics courses at the 4000-level	1.0
12. 1.0 credit from:		1.0
ERTH 4908 [1.0]	Honours Thesis	
OR		
GEOG 4005 [0.5]	Directed Studies in Geography and 0.5 credit in EARTH, GEOG or GEOM at the 4000-level	
OR		
GEOG 4906 [1.0]	Honours Research Project	
B. Credits Not Included in the Major CGPA (7.0 credits)		
13. 1.0 credit in:		1.0
MATH 1007 [0.5]	Elementary Calculus I	
MATH 1107 [0.5]	Linear Algebra I	
14. 1.0 credit in:		1.0
CHEM 1001 [0.5]	General Chemistry I	
& CHEM 1002 [0.5]	General Chemistry II	
15. 1.0 credit in:		1.0
PHYS 1007 [0.5]	Elementary University Physics I	
& PHYS 1008 [0.5]	Elementary University Physics II	
16. 0.5 credit from:		0.5
GEOG 2006 [0.5]	Introduction to Quantitative Research	
STAT 2507 [0.5]	Introduction to Statistical Modeling I	
17. 0.5 credit in:		0.5
COMP 1005 [0.5]	Introduction to Computer Science I	
18. 0.5 credit in	approved electives (see list below)	0.5
19. 0.5 credit in:		0.5
ISAP 1000 [0.5]	Seminar in Science (or approved course outside of the faculties of Science and Engineering and Design)	
20. 1.5 credits in	approved courses outside of the faculties of Science and Engineering and Design	1.5
21. 0.5 credit in	free elective	0.5
Total Credits		20.0

Approved Electives - B.Sc. Earth Sciences and Physical Geography

Biology

BIOL 1103 [0.5]	Foundations of Biology I	
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BIOL 1104 [0.5]	Foundations of Biology II	
Computer Science		
COMP 1006 [0.5]	Introduction to Computer Science II	
Chemistry		
CHEM 2103 [0.5]	Physical Chemistry I	
CHEM 2203 [0.5]	Organic Chemistry I	
CHEM 2207 [0.5]	Introduction to Organic Chemistry I	
CHEM 2501 [0.5]	Introduction to Inorganic and Bioinorganic Chemistry	
Mathematics		
MATH 1005 [0.5]	Differential Equations and Infinite Series for Engineering or Physics	
MATH 2007 [0.5]	Elementary Calculus II	
MATH 2107 [0.5]	Linear Algebra II	
Physics		
PHYS 2202 [0.5]	Wave Motion and Optics	
Statistics		
STAT 2509 [0.5]	Introduction to Statistical Modeling II	
Physical Geography		
B.Sc. Honours (20.0 credits)		
A. Credits Included in the Major CGPA (10.0 credits)		
1. 1.0 credit from:		1.0
GEOG 1010 [0.5]	Global Environmental Systems	
or EARTH 1002 [0.5]	The Earth and Life Odyssey: A Journey Through Billions of Years	
and		
GEOM 1004 [0.5]	Maps, Satellites and the Geospatial Revolution	
2. 1.0 credit in:		1.0
GEOG 2013 [0.5]	Weather and Water	
GEOG 2014 [0.5]	The Earth's Surface	
3. 0.5 credit from:		0.5
GEOG 2006 [0.5]	Introduction to Quantitative Research	
STAT 2507 [0.5]	Introduction to Statistical Modeling I	
4. 0.5 credit from:		0.5
GEOG 3000 [0.5]	Honours Field Course	
GEOG 3010 [0.5]	Field Methods in Physical Geography	
5. 2.5 credits from:		2.5
GEOG 3003 [0.5]	Quantitative Geography	
GEOG 3102 [0.5]	Geomorphology	
GEOG 3103 [0.5]	Watershed Hydrology	
GEOG 3104 [0.5]	Principles of Biogeography	
GEOG 3105 [0.5]	Climate and Atmospheric Change	
GEOG 3106 [0.5]	Aquatic Science and Management	
GEOG 3108 [0.5]	Soil Properties	
GEOM 3002 [0.5]	Introduction to Remote Sensing	
6. 1.5 credits from:		1.5
GEOG 3000 [0.5]	Honours Field Course	
GEOG 3003 [0.5]	Quantitative Geography	
GEOG 3010 [0.5]	Field Methods in Physical Geography	
GEOG 3102 [0.5]	Geomorphology	
GEOG 3103 [0.5]	Watershed Hydrology	
GEOG 3104 [0.5]	Principles of Biogeography	

GEOG 3105 [0.5]	Climate and Atmospheric Change	
GEOG 3108 [0.5]	Soil Properties	
GEOG 4000 [0.5]	Field Studies	
GEOG 4005 [0.5]	Directed Studies in Geography	
GEOG 4013 [0.5]	Cold Region Hydrology	
GEOG 4017 [0.5]	Global Biogeochemical Cycles	
GEOG 4101 [0.5]	Two Million Years of Environmental Change	
GEOG 4103 [0.5]	Water Resources Engineering	
GEOG 4104 [0.5]	Microclimatology	
GEOG 4108 [0.5]	Permafrost	
7. 2.0 credits from:		2.0
GEOM at the 4000 level or		
GEOG 4000 [0.5]	Field Studies	
GEOG 4004 [0.5]	Environmental Impact Assessment	
GEOG 4005 [0.5]	Directed Studies in Geography	
GEOG 4013 [0.5]	Cold Region Hydrology	
GEOG 4017 [0.5]	Global Biogeochemical Cycles	
GEOG 4101 [0.5]	Two Million Years of Environmental Change	
GEOG 4103 [0.5]	Water Resources Engineering	
GEOG 4104 [0.5]	Microclimatology	
GEOG 4108 [0.5]	Permafrost	
GEOG 4406 [0.5]	Practicum I	
GEOG 4408 [0.5]	Practicum II	
8. 1.0 credit in:		1.0
GEOG 4906 [1.0]	Honours Research Project	
B. Credits Not Included in the Major CGPA (10.0 credits)		
9. 1.0 credit in	Experimental Science Electives	1.0
10. 0.5 credit in:		0.5
MATH 1007 [0.5]	Elementary Calculus I	
11. 0.5 credit in	MATH or COMP	0.5
12. 2.0 credits in	Science Continuation, not in GEOG	2.0
13. 1.0 credits in	Science Faculty Electives	1.0
14. 0.5 credit from:		0.5
ISAP 1000 [0.5]	Seminar in Science (or approved courses outside the faculties of Science and Engineering and Design)	
15. 1.0 credit in	approved courses outside the faculties of Science and Engineering and Design, not in GEOG	1.0
16. 0.5 credit in	approved courses outside the faculties of Science and Engineering and Design	0.5
17. 3.0 credits in	free electives.	3.0
Total Credits		20.0

Bachelor of Global and International Studies (B.G.In.S.)

Note: Details regarding graduation requirements, the international experience requirement, and the language requirement for the B.G.In.S. degree can be found at the **B.G.In.S. program page**.

Specialization in Globalization and the Environment

B.G.In.S. Honours (20.0 credits)

A. Credits Included in the Major CGPA (12.0 credits)

1. 4.5 credits in:	4.5
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GINS 1000 [0.5]	Global History
GINS 1010 [0.5]	International Law and Politics
GINS 1020 [0.5]	Ethnography, Globalization and Culture
GINS 2000 [0.5]	Ethics and Globalization
GINS 2010 [0.5]	Globalization and International Economic Issues
GINS 2020 [0.5]	Global Literatures
GINS 3010 [0.5]	Global and International Theory
GINS 3020 [0.5]	Places, Boundaries, Movements and Global Environmental Change
GINS 4090 [0.5]	Honours Seminar in Global and International Studies

2. 0.0 credit in: International Experience Requirement Preparation

GINS 1300 [0.0]	International Experience Requirement Preparation
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3. 7.5 credits in: the Specialization

a. 0.5 credit from:	Foundations I	0.5
ENST 1000 [0.5]	Introduction to Environmental and Climate Change Studies	

OR

GEOG 1020/ENST 1020 [0.5]	People, Places and Environments	
b. 1.0 credit in:	Foundations II	1.0
GEOG 1010 [0.5]	Global Environmental Systems	
GEOG 2200 [0.5]	Global Connections	

c. 1.5 credits from: Globalization **1.5**

GEOG 2023 [0.5]	Cities, Inequality and Urban Change
GEOG 2300 [0.5]	Space, Place and Culture
GEOG 3023 [0.5]	Cities in a Global World
GEOG 3024 [0.5]	Understanding Globalization
GEOG 3025 [0.5]	Geographies of Selected Regions
GEOG 3030 [0.5]	Regional Field Excursion
GEOG 3404 [0.5]	Geographies of Economic Development

d. 2.0 credits from: Global Environment **2.0**

ANTH 3355 [0.5]	Anthropology and the Environment
GEOG 2500/ENST 2500 [0.5]	Climate Change: Social Science Perspectives
GEOG 3022/ENST 3022 [0.5]	Environmental and Natural Resources
GEOG 3206 [0.5]	Health, Environment, and Society
GEOG 3209 [0.5]	Sustainability and Environment in the South

HRSJ 3503 [0.5]	Global Environmental Justice
PSCI 3801 [0.5]	Environmental Politics
TSES 3002 [0.5]	Energy and Sustainability

e. 1.0 credit in: Research Methodologies **1.0**

GEOG 2005/ENST 2005 [0.5]	Introduction to Qualitative Research
GEOG 2006/ENST 2006 [0.5]	Introduction to Quantitative Research

f. 1.5 credits from: Honours Seminars **1.5**

GEOG 4005/ENST 4005 [0.5]	Directed Studies in Geography (topic in Global Environmental Issues)
GEOG 4022 [0.5]	Seminar in People, Resources and Environmental Change

GEOG 4023 [0.5]	Seminar in Special Topics on the City	
GEOG 4024 [0.5]	Seminar in Globalization	
GEOG 4909 [1.0]	Honours Research Thesis (topic in Globalization and the Environment)	
PSCI 4808 [0.5]	Global Environmental Politics	
B. Credits Not Included in the Major CGPA (8.0 credits)		
4. 8.0 credits in:	free electives	8.0
C. Additional Requirements		
5.	The International Experience requirement must be met.	
6.	The Language requirement must be met.	
Total Credits		20.0

Stream in Globalization and the Environment B.G.In.S. (15.0 credits)

A. Credits Included in the Major CGPA (8.0 credits)		
1. 4.0 credits in:	Core Courses	4.0
GINS 1000 [0.5]	Global History	
GINS 1010 [0.5]	International Law and Politics	
GINS 1020 [0.5]	Ethnography, Globalization and Culture	
GINS 2000 [0.5]	Ethics and Globalization	
GINS 2010 [0.5]	Globalization and International Economic Issues	
GINS 2020 [0.5]	Global Literatures	
GINS 3010 [0.5]	Global and International Theory	
GINS 3020 [0.5]	Places, Boundaries, Movements and Global Environmental Change	
2. 4.0 credits from:	the Stream	4.0
a. Foundations		
GEOG 1010 [0.5]	Global Environmental Systems	
GEOG 1020/ENST 1020 [0.5]	People, Places and Environments	
GEOG 2200 [0.5]	Global Connections	
b. Globalization		
GEOG 2023 [0.5]	Cities, Inequality and Urban Change	
GEOG 2300 [0.5]	Space, Place and Culture	
GEOG 3023 [0.5]	Cities in a Global World	
GEOG 3024 [0.5]	Understanding Globalization	
GEOG 3025 [0.5]	Geographies of Selected Regions	
GEOG 3404 [0.5]	Geographies of Economic Development	
c. Global Environment		
ANTH 3355 [0.5]	Anthropology and the Environment	
GEOG 2500/ENST 2500 [0.5]	Climate Change: Social Science Perspectives	
GEOG 3022/ENST 3022 [0.5]	Environmental and Natural Resources	
GEOG 3206 [0.5]	Health, Environment, and Society	
GEOG 3209 [0.5]	Sustainability and Environment in the South	
HRSJ 3503 [0.5]	Global Environmental Justice	
PSCI 3801 [0.5]	Environmental Politics	
TSES 3002 [0.5]	Energy and Sustainability	
d. Research Methodologies		
GEOG 2005/ENST 2005 [0.5]	Introduction to Qualitative Research	

GEOG 2006/ENST 2006 [0.5]	Introduction to Quantitative Research	
B. Credits Not Included in the Major CGPA (7.0 credits):		
3. 7.0 credits in:	Free Electives	7.0
C. Additional Requirements		
4.	The Language requirement must be met.	
Total Credits		15.0

Minor in Geography (4.0 credits)

Open to all undergraduate degree students not in Geography programs or the B.G.In.S. Specialization or Stream in Globalization and the Environment.

Students are required to present a Minor CGPA of 4.00 or higher at graduation in order to be awarded a Minor in Geography.

Requirements:		
1. 1.0 credit in:		1.0
GEOG 1010 [0.5]	Global Environmental Systems	
GEOG 1020 [0.5]	People, Places and Environments	
2. 0.5 credit from:		0.5
GEOG 2013 [0.5]	Weather and Water	
GEOG 2014 [0.5]	The Earth's Surface	
GEOG 2020 [0.5]	Ecosystems of Canada	
3. 0.5 credit from:		0.5
GEOG 2005 [0.5]	Introduction to Qualitative Research	
GEOG 2006 [0.5]	Introduction to Quantitative Research	
GEOM 1004 [0.5]	Maps, Satellites and the Geospatial Revolution	
4. 0.5 credit from:		0.5
GEOG 2023 [0.5]	Cities, Inequality and Urban Change	
GEOG 2200 [0.5]	Global Connections	
GEOG 2300 [0.5]	Space, Place and Culture	
GEOG 2500 [0.5]	Climate Change: Social Science Perspectives	
5. 1.0 credit in	GEOG and/or GEOM at the 3000-level or above	1.0
6. 0.5 credit in	GEOG or GEOM	0.5
7.	The remaining requirements of the major discipline(s) and degree must be satisfied.	
Total Credits		4.0

Minor in Physical Geography (4.0 credits)

Open to all undergraduate degree students not in Geography programs.

Students are required to present a Minor CGPA of 4.00 or higher at graduation in order to be awarded a Minor in Physical Geography.

Requirements:		
1. 0.5 credit from:		0.5
GEOG 1010 [0.5]	Global Environmental Systems	
2. 1.0 credit in:		1.0
GEOG 2013 [0.5]	Weather and Water	

GEOG 2014 [0.5]	The Earth's Surface	2.5
3. 2.5 credits from:		
GEOM 3002 [0.5]	Introduction to Remote Sensing	
GEOG 3003 [0.5]	Quantitative Geography	
GEOG 3102 [0.5]	Geomorphology	
GEOG 3103 [0.5]	Watershed Hydrology	
GEOG 3104 [0.5]	Principles of Biogeography	
GEOG 3105 [0.5]	Climate and Atmospheric Change	
GEOG 3106 [0.5]	Aquatic Science and Management	
GEOG 3108 [0.5]	Soil Properties	
GEOG 4013 [0.5]	Cold Region Hydrology	
GEOG 4017 [0.5]	Global Biogeochemical Cycles	
GEOG 4101 [0.5]	Two Million Years of Environmental Change	
GEOG 4104 [0.5]	Microclimatology	
GEOG 4108 [0.5]	Permafrost	
4. The remaining requirements of the major discipline(s) and degree must be satisfied.		
Total Credits		4.0

Minor in Urban Studies (4.0 credits)

Only students pursuing an undergraduate program (except the BA Honours in Geography with a Concentration in Urban Geography) requiring at least 20.0 credits to graduate may be admitted to the Urban Studies minor.

Students are required to present a Minor CGPA of 4.00 or higher at graduation in order to be awarded a Minor in Urban Studies.

Requirements:

1. 1.0 credit from:			1.0
FYSM 1107 [1.0]	Social Justice and the City		
GEOG 1020 [0.5]	People, Places and Environments		
GEOG 1023 [0.5]	Introduction to Cities and Urbanization		
GEOM 1004 [0.5]	Maps, Satellites and the Geospatial Revolution		
2. 1.0 credit from:			1.0
GEOG 2023 [0.5]	Cities, Inequality and Urban Change		
GEOG 2200 [0.5]	Global Connections		
GEOM 2007 [0.5]	Vector GIS: Points, Lines and Polygons		
3. 0.5 credit in:			0.5
GEOG 3023 [0.5]	Cities in a Global World		
4. 0.5 credit from:			0.5
AFRI 3004 [0.5]	The African City		
ARCU 3100 [0.5]	The Morphology of the City		
HRSJ 3002 [0.5]	Right to the City		
5. 0.5 credit from:			0.5
GEOG 4023 [0.5]	Seminar in Special Topics on the City		
GEOG 4323 [0.5]	Urban and Regional Planning		
6. 0.5 credit from:			0.5
ARCH 4201 [0.5]	History of Modern Housing		
ARCU 4103 [0.5]	Cities		
ARCU 4300 [0.5]	Theories of Urbanism		
ARCU 4700 [0.5]	Urban Utopias		
ARCU 4801 [0.5]	Topics in Urbanism		

GEOG 4000 [0.5]	Field Studies (when offered with an urban theme)	
GEOG 4005 [0.5]	Directed Studies in Geography (with urban theme)	
GEOG 4007 [0.5]	Special Topics in Geography and Environmental Studies	
INDG 4001 [0.5]	Indigenous Urbanisms	
7. The remaining requirements of the major discipline(s) and degree must be satisfied.		
Total Credits		4.0

B.A. Regulations

The regulations presented below apply to all Bachelor of Arts programs. In addition to the requirements presented here, students must satisfy the University regulations common to all undergraduate students including the process of Academic Continuation Evaluation (consult the *Academic Regulations of the University* section of this Calendar).

First-Year Seminars

B.A. degree students are strongly encouraged to include a First-Year Seminar (FYSM) during their first 4.0 credits of registration. Students are limited to 1.0 credit in FYSM and can only register in a FYSM while they have first-year standing in their B.A. program.

Breadth Requirement

Among the credits presented at graduation, students in both the B.A. and the B.A. Honours degrees and B.Co.M.S. are required to include 3.0 breadth credits, which must include 1.0 credit in three of the four breadth areas identified below. Credits that fulfil requirements in the Major, Minor, Concentration, Specialization, or Stream may also be used to fulfil the Breadth Requirement.

Students admitted with a completed university degree are exempt from breadth requirements.

Students in the following interdisciplinary programs are exempt from the B.A. breadth requirement.

- African Studies
- Criminology and Criminal Justice
- Environmental Studies
- Human Rights
- Human Rights and Social Justice

Breadth Area 1: Culture and Communication

American Sign Language, Art History, Art and Culture, Communication and Media Studies, Digital Humanities, English, Film Studies, French, Journalism, Media Production and Design, Music, and Languages (Arabic, English as a Second Language, German, Greek, Hebrew, Indigenous Languages, Italian, Japanese, Korean, Latin, Mandarin, Portuguese, Russian, Spanish)

Subject codes: ARAB, ARTH, ASLA, CHIN, COMS, DIGH, ENGL, ESLA, FILM, FINS, FREN, GERM, GREK, HEBR, ITAL, JAPA, JOUR, KORE, LANG, LATN, MPAD, MUSI, PORT, RUSS, SPAN

Breadth Area 2: Humanities

African Studies, Applied Linguistics and Discourse Studies, Archaeology, Canadian Studies, Child Studies, Classical Civilization, Critical Race Studies, Directed Interdisciplinary Studies, Disability Studies, Environmental and Climate Humanities, European and Russian Studies, History, Human Rights and Social Justice, Humanities, Indigenous Studies, Latin American and Caribbean Studies, Linguistics, Medieval and Early Modern Studies, Philosophy, Religion, Sexuality Studies, South Asian Studies, and Women's and Gender Studies.

Subject codes: AFRI, ALDS, ARCY, CDNS, CHST, CLCV, CRST, DBST, DIST, EACH, EURR, HIST, HRSJ, HUMR, HUMS, INDG, LACS, LING, MEMS, PHIL, RELI, SAST, SXST, WGST

Breadth Area 3: Science, Engineering, and Design

Architecture, Biology, Biotechnology, Chemistry, Computer Science, Data Science, Earth Sciences, Engineering, Environmental Science, Food Science and Nutrition, Health Sciences, Industrial Design, Information Resource Management, Information Technology (BIT), Information Technology (ITEC), Interactive Multimedia and Design, Interdisciplinary Science and Practice, Mathematics, Neuroscience, Network Technology, Nursing, Optical Systems and Sensors, Photonics, Statistics, Physics, and Technology, Society, Environment.

Subject codes: ACSE, AERO, ARCC, ARCH, ARCN, ARCS, ARCU, BIOC, BIOL, BIT, BTEC, CHEM, CIVE, COMP, CSEC, DATA, ECOR, ELEC, ENSC, ENVE, EARTH, FOOD, HLTH, IDES, IMD, IRM, ISAP, ISCI, ISCS, ISYS, ITEC, MAAE, MATH, MECH, MECT, NET, NEUR, NSCI, NURS, OSS, PHYS, PLT, SREE, STAT, SYSC, TSES

Breadth Area 4: Social Sciences

Anthropology, Business, Cognitive Science, Criminology and Criminal Justice, Economics, Environmental Studies, Geography, Geomatics, Global and International Studies, Global Politics, Interdisciplinary Public Affairs, International Affairs, Law, Migration and Diaspora Studies, Political Management, Political Science, Psychology, Public Administration, Public Affairs and Policy Management, Social Work, Sociology/Anthropology, Sociology

Subject codes: ANTH, BUSI, CGSC, CRCJ, ECON, ENST, GEOG, GEOM, GINS, GPOL, INAF, IPAF, LAWS, MGDS, PADM, PAPM, POLM, PSCI, PSYC, SOCI, SOWK

Declared and Undeclared Students

Degree students are considered "Undeclared" if they have been admitted to a degree, but have not yet selected and been accepted into a program within that degree. The status "Undeclared" is available only in the B.A. and B.Sc. degrees. Undeclared students must apply to enter a program upon or before completing 3.5 credits.

Change of Program Within the B.A. Degree

To transfer to a program within the B.A. degree, applicants must normally be *Eligible to Continue* (EC) in the new program, by meeting the CGPA thresholds described

in Section 3.1.9 of the *Academic Regulations of the University*.

Applications to declare or change programs within the B.A. degree online must be made online through Carleton Central by completing a Change of Program Elements (COPE) application form within the published deadlines. Acceptance into a program, or into a program element or option, is subject to any enrollment limitations, as well as specific program, program element, or option requirements as published in the relevant Calendar entry.

Minors, Concentrations, and Specializations

Students may add a Minor, Concentration, or Specialization by completing a Change of Program Elements (COPE) application form online through Carleton Central. Acceptance into a Minor, Concentration, or Specialization normally requires that the student be *Eligible to Continue* (EC) and is subject to any specific requirements of the intended Minor, Concentration, or Specialization as published in the relevant Calendar entry and in Section 3.1.9 of the *Academic Regulations of the University*.

Mention : français

Students registered in certain B.A. programs may earn the diploma notation *Mention : français* by completing part of their program requirements in French, and by demonstrating knowledge of the history and culture of French Canada. The general requirements are listed below. For more specific details, consult the departmental program entries.

Students in a B.A. Honours program must present:

1. 1.0 credit in French language;
2. 1.0 credit devoted to the history and culture of French Canada;
3. 1.0 credit at the 2000- or 3000-level in the Honours discipline taken in French; and
4. 1.0 credit at the 4000-level in the Honours discipline taken in French.

Students in a B.A. program must present:

1. 1.0 credit in advanced French;
2. 1.0 credit devoted to the history and culture of French Canada;
3. 1.0 credit at the 2000- or 3000-level in the Major discipline taken in French.

Students in Combined Honours programs must fulfil the *Mention : français* requirement in both disciplines.

Courses taught in French (Items 3 and 4, above) may be taken at Carleton, at the University of Ottawa on the Exchange Agreement, or at a francophone university on a Letter of Permission. Students planning to take courses on exchange or on a Letter of Permission should take careful note of the residence requirement for a minimum number of Carleton courses in their programs. Consult the *Academic Regulations of the University* section of this Calendar for information regarding study on exchange or Letter of Permission.

B.Sc. Regulations

The regulations presented in this section apply to all Bachelor of Science programs. In addition to the requirements presented here, students must satisfy the University regulations common to all undergraduate students including the process of Academic Continuation Evaluation (see the *Academic Regulations of the University* section of this Calendar).

Breadth Requirement for the B.Sc.

Students in a Bachelor of Science program must present the following credits at graduation:

1. 2.0 credits in Science Continuation courses not in the major discipline; **students completing a double major are considered to have completed this requirement providing they have 2.0 credits in Science Continuation courses in each of the two majors;**
2. 2.0 credits in courses outside of the faculties of Science and Engineering and Design (may include ISAP 1000)

In most cases, the requirements for individual B.Sc. programs, as stated in this Calendar, contain these requirements, explicitly or implicitly.

Students admitted to B.Sc. programs by transfer from another institution must present at graduation (whether taken at Carleton or elsewhere):

1. 2.0 credits in courses outside of the faculties of Science and Engineering and Design (may include ISAP 1000) if the student received fewer than 10.0 transfer credits; or,
2. 1.0 credit in courses outside of the faculties of Science and Engineering and Design (may include ISAP 1000) if the student received 10.0 or more transfer credits.

Declared and Undeclared Students

Degree students are considered "Undeclared" if they have been admitted to a degree, but have not yet selected and been accepted into a program within that degree. The status "Undeclared" is available only in the B.A. and B.Sc. degrees. Undeclared students must apply to enter a program upon or before completing 3.5 credits.

Change of Program within the B.Sc. Degree

To transfer to a program within the B.Sc. degree, applicants must normally be *Eligible to Continue* (EC) in the new program, by meeting the CGPA thresholds described in Section 3.1.9 of the *Academic Regulations of the University*.

Applications to declare or change programs within the B.Sc. degree must be made online through Carleton Central by completing a Change of Program Elements (COPE) application form within the published deadlines. Acceptance into a program, or into a program element or option, is subject to any enrolment limitations, and/or specific program, program element or option requirements as published in the relevant Calendar entry.

Minors, Concentrations, and Specializations

Students may add a Minor, Concentration, or Specialization by completing a Change of Program Elements (COPE) application form online through Carleton Central. Acceptance into a Minor, Concentration, or Specialization normally requires that the student be *Eligible to Continue* (EC) and is meeting the minimum CGPAs described in Section 3.1.9 of the *Academic Regulations of the University*, as well as being subject to any specific requirements of the intended Minor, Concentration, or Specialization as published in the relevant Calendar entry.

Experimental Science Requirement

Students in a B.Sc. degree program must present at graduation at least two full credits of Experimental Science chosen from two different departments or institutes from the list below:

Approved Experimental Science Courses

Biochemistry	
BIOC 2200 [0.5]	Cellular Biochemistry
BIOC 3103 [0.5]	Experimental Biochemistry I: Principles and Practices
BIOC 3104 [0.5]	Experimental Biochemistry II: Research Experience
BIOC 4001 [0.5]	Methods in Biochemistry
BIOC 4201 [0.5]	Advanced Cell Culture and Tissue Engineering
Biology	
BIOL 1103 [0.5]	Foundations of Biology I
BIOL 1104 [0.5]	Foundations of Biology II
BIOL 2001 [0.5]	Animals: Form and Function
BIOL 2002 [0.5]	Plants: Form and Function
BIOL 2104 [0.5]	Introductory Genetics
BIOL 2200 [0.5]	Cellular Biochemistry
BIOL 2600 [0.5]	Ecology
Chemistry	
CHEM 1001 [0.5]	General Chemistry I
CHEM 1002 [0.5]	General Chemistry II
CHEM 2103 [0.5]	Physical Chemistry I
CHEM 2203 [0.5]	Organic Chemistry I
CHEM 2204 [0.5]	Organic Chemistry II
CHEM 2302 [0.5]	Analytical Chemistry I
CHEM 2303 [0.5]	Analytical Chemistry II
CHEM 2800 [0.5]	Foundations for Environmental Chemistry
Earth Sciences	
ERTH 1002 [0.5]	The Earth and Life Odyssey: A Journey Through Billions of Years
ERTH 2102 [0.5]	Mineralogy to Petrology
ERTH 2404 [0.5]	Engineering Geoscience
ERTH 2802 [0.5]	Field Geology I
ERTH 3111 [0.5]	Vertebrate Evolution: Mammals, Reptiles, and Birds
ERTH 3112 [0.5]	Vertebrate Evolution: Fish and Amphibians
ERTH 3204 [0.5]	Mineral Deposits
ERTH 3205 [0.5]	Physical Hydrogeology
Food Sciences	

FOOD 3001 [0.5]	Food Chemistry
FOOD 3002 [0.5]	Food Analysis
FOOD 3005 [0.5]	Food Microbiology

Geography

GEOG 1010 [0.5]	Global Environmental Systems
GEOG 3108 [0.5]	Soil Properties

Neuroscience

NEUR 3206 [0.5]	Sensory and Motor Neuroscience
NEUR 3207 [0.5]	Systems Neuroscience
NEUR 4600 [0.5]	Advanced Lab in Neuroanatomy

Physics

PHYS 1001 [0.5]	Foundations of Physics I
PHYS 1002 [0.5]	Foundations of Physics II
PHYS 1003 [0.5]	Introductory Mechanics and Thermodynamics
PHYS 1004 [0.5]	Introductory Electromagnetism and Wave Motion
PHYS 1007 [0.5]	Elementary University Physics I
PHYS 1008 [0.5]	Elementary University Physics II
PHYS 2007 [0.5]	Second Year Physics Laboratory: Selected Experiments and Seminars
PHYS 3007 [0.5]	Third Year Physics Laboratory: Selected Experiments and Seminars
PHYS 3606 [0.5]	Modern Physics II
PHYS 3608 [0.5]	Modern Applied Physics

Course Categories for B.Sc. Programs

Science Geography Courses

GEOG 1010 [0.5]	Global Environmental Systems
GEOG 2006 [0.5]	Introduction to Quantitative Research
GEOG 2013 [0.5]	Weather and Water
GEOG 2014 [0.5]	The Earth's Surface
GEOG 3003 [0.5]	Quantitative Geography
GEOG 3010 [0.5]	Field Methods in Physical Geography
GEOG 3102 [0.5]	Geomorphology
GEOG 3103 [0.5]	Watershed Hydrology
GEOG 3104 [0.5]	Principles of Biogeography
GEOG 3105 [0.5]	Climate and Atmospheric Change
GEOG 3106 [0.5]	Aquatic Science and Management
GEOG 3108 [0.5]	Soil Properties
GEOG 4000 [0.5]	Field Studies
GEOG 4005 [0.5]	Directed Studies in Geography
GEOG 4013 [0.5]	Cold Region Hydrology
GEOG 4017 [0.5]	Global Biogeochemical Cycles
GEOG 4101 [0.5]	Two Million Years of Environmental Change
GEOG 4103 [0.5]	Water Resources Engineering
GEOG 4104 [0.5]	Microclimatology
GEOG 4108 [0.5]	Permafrost

Science Psychology Courses

PSYC 2001 [0.5]	Introduction to Research Methods in Psychology
PSYC 2002 [0.5]	Introduction to Statistics in Psychology

PSYC 2700 [0.5]	Introduction to Cognitive Psychology
PSYC 3000 [1.0]	Design and Analysis in Psychological Research
PSYC 3506 [0.5]	Cognitive Development
PSYC 3700 [1.0]	Cognition (Honours Seminar)
PSYC 3702 [0.5]	Perception
PSYC 2307 [0.5]	Human Neuropsychology I
PSYC 3307 [0.5]	Human Neuropsychology II

Science Continuation Courses

A course at the 2000 level or above may be used as a Science Continuation credit in a B.Sc. program if it is not in the student's major discipline, and is chosen from the following:

BIOC (Biochemistry)
BIOL (Biology) except BIOL 4810 which may be used only as a free elective for any B.Sc. program. Biochemistry students may use BIOL 2005 only as a free elective.
CHEM (Chemistry)
COMP (Computer Science) A maximum of two half-credits at the 1000-level in COMP, excluding COMP 1001 may be used as Science Continuation credits.
ERTH (Earth Sciences), except ERTH 2415 which may be used only as a free elective for any B.Sc. program. Students in Earth Sciences programs may use ERTH 2401, ERTH 2402, and ERTH 2403 only as free electives.
Engineering. Students wishing to register in Engineering courses must obtain the permission of the Faculty of Engineering and Design.
ENSC (Environmental Science)
FOOD (Food Science and Nutrition)
GEOM (Geomatics)
HLTH (Health Sciences)
ISAP (Interdisciplinary Science Practice)
MATH (Mathematics)
NEUR (Neuroscience)
PHYS (Physics), except PHYS 2903
Science Geography Courses (see list above)
Science Psychology Courses (see list above)
STAT (Statistics)
TSES (Technology, Society, Environment) except TSES 2305. Biology students may use these courses only as free electives. Integrated Science and Environmental Science students may include these courses in their programs but may not count them as part of the Science Sequence.

Science Faculty Electives

Science Faculty Electives are courses at the 1000-4000 level chosen from:

BIOC (Biochemistry) except BIOC 1900 which may be used only as a free elective for any B.Sc. program.
BIOL (Biology) except BIOL 4810 which may be used only as a free elective for any B.Sc. program. Biology & Biochemistry students may use BIOL 1010 and BIOL 2005 only as free electives.
CHEM (Chemistry) except CHEM 1003, CHEM 1004 and CHEM 1007

COMP (Computer Science) except COMP 1001
ERTH (Earth Sciences) except ERTH 1004 and ERTH 2415. Earth Sciences students may use ERTH 2401, ERTH 2402 and ERTH 2403 only as free electives.

Engineering

ENSC (Environmental Science)

FOOD (Food Science and Nutrition)

GEOM (Geomatics)

HLTH (Health Science)

ISAP (Interdisciplinary Science Practice)

MATH (Mathematics)

NEUR (Neuroscience)

PHYS (Physics) except PHYS 1901, PHYS 1902, PHYS 1905, and PHYS 2903

Science Geography (see list above)

Science Psychology (see list above)

STAT (Statistics)

TSES (Technology, Society, Environment) Biology students may use these courses only as free electives.

Advanced Science Faculty Electives

Advanced Science Faculty Electives are courses at the 2000-4000 level chosen from the Science Faculty Electives list above.

Approved Courses Outside the Faculties of Science and Engineering and Design (may include ISAP 1000)

All courses offered by the Faculty of Arts and Social Sciences, the Faculty of Public and Global Affairs, and the Sprott School of Business are approved as Arts or Social Sciences courses EXCEPT FOR: All Science Geography courses (see list above), all Geomatics (GEOM) courses, all Science Psychology courses (see list above). ISAP 1000 may be used as an Approved Course Outside the Faculties of Science and Engineering and Design.

Free Electives

Any course is allowable as a Free Elective providing it is not prohibited (see below). Students are expected to comply with prerequisite requirements and enrolment restrictions for all courses as published in this Calendar.

Courses Allowable Only as Free Electives in any B.Sc. Program

BIOL 4810 [0.5]	Education Research in Undergraduate Science
CHEM 1003 [0.5]	The Chemistry of Food, Health and Drugs
CHEM 1004 [0.5]	Drugs and the Human Body
CHEM 1007 [0.5]	Chemistry of Art and Artifacts
ERTH 1004 [0.5]	Earth's Epic Tale: A Story Across Billions of Years
ERTH 2415 [0.5]	Natural Disasters
ISCI 1001 [0.5]	Introduction to the Environment
ISCI 2000 [0.5]	Natural Laws
ISCI 2002 [0.5]	Human Impacts on the Environment
PHYS 1901 [0.5]	Planetary Astronomy
PHYS 1902 [0.5]	From our Star to the Cosmos
PHYS 1905 [0.5]	Physics Behind Everyday Life
PHYS 2903 [0.5]	Physics Towards the Future

Prohibited Courses

The following courses are not acceptable for credit in any B.Sc. program:

COMP 1001 [0.5]	Introduction to Computational Thinking for Arts and Social Science Students
MATH 1009 [0.5]	Mathematics for Business
MATH 1119 [0.5]	Linear Algebra: with Applications to Business
MATH 1401 [0.5]	Elementary Mathematics for Economics I
MATH 1402 [0.5]	Elementary Mathematics for Economics II
all 0000-level courses	

Co-operative Education

For more information about how to apply for the Co-op program and how the Co-op program works please visit the Co-op website.

All students participating in the Co-op program are governed by the Undergraduate Co-operative Education Policy.

Undergraduate Co-operative Education Policy Admission Requirements

Students can apply to Co-op in one of two ways: directly from high school, or after beginning a degree program at Carleton.

If a student applies to a degree program with a Co-op option from high school, their university grades will be reviewed two terms to one year prior to their first work term to ensure they meet the academic requirements after their first or second year of study. The time at which the evaluation takes place depends on the program of study. Students will automatically receive an admission decision via their Carleton email account.

Students who did not request Co-op at the time they applied to Carleton can request Co-op after they begin their university studies. To view application instructions and deadlines, please visit carleton.ca/co-op.

To be admitted to Co-op, a student must successfully complete 5.0 or more credits that count towards their degree, meet the minimum CGPA requirement(s) for the student's Co-op option, and fulfil any specified course prerequisites. To see the unique admission and continuation requirements for each Co-op option, please refer to the specific degree programs listed in the Undergraduate Calendar.

Participation Requirements

Co-op Participation Agreement

All students must adhere to the policies found within the Co-op Participation Agreement.

COOP 1000

Once a student has been admitted to the Co-op Program, they will be given access to register in COOP 1000. This zero-credit online course must be completed at least two terms prior to the student's first work term.

Communication with the Co-op Office

Students must maintain contact with the Co-op Office during their job search and while on a work term. All email communication will be conducted via the students' Carleton email account.

Employment

Although every effort is made to ensure a sufficient number of job postings for all Co-op students, no guarantee of employment can be made. The Co-op job search process is competitive, and success is dependent upon factors such as current market conditions, academic performance, skills, motivation, and level of commitment to the job search. It is the student's responsibility to apply for positions via the Co-op job board in addition to actively conducting a self-directed job search. Students who do not obtain a co-op work term are expected to continue with their academic studies. It should be noted that hiring priority for positions within the Federal Government of Canada is given to Canadian citizens.

Registration

- Students must be registered as full-time during all fall and winter study terms beginning the term in which they enroll in COOP 1000.
- Students will be registered in a Co-op Work Term course while at work. This course does not carry academic course credit, but is noted on academic transcripts.
- Students may register in a 0.5 credit during a work term, provided the course is offered during the evening or is offered asynchronously online.
- Students must have at least one term of full-time studies left to complete following their final co-op work term. Students cannot end their degree on a work term.

Work Term Assessment and Evaluation

Work Term Evaluation

Employers are responsible for submitting to Carleton University final performance evaluations for their Co-op students at the end of their work terms.

Work Term Assessment

In order to successfully complete the co-op work term, students must receive a Satisfactory (SAT) grade on their Co-op Work Term Report, which they must submit at the completion of each four-month work term.

Graduation with the Co-op Designation

In order to graduate with the Co-op Designation, students must satisfy all requirements of the degree program in addition to the successful completion of three or four work terms (the number is dependent upon the student's academic program). Students found in violation of the Co-op Participation Agreement may have the Co-op Designation withheld.

Note: Participation in the co-op option will add up to one additional year for a student to complete their degree program.

Voluntary Withdrawal from the Co-op Option

Students who are currently on a co-op work term or who have already committed to a co-op work term either verbally or in writing may not leave the position and/or withdraw from the co-op option until they have completed the work term and all related requirements.

Involuntary or Required Withdrawal from the Co-op Option

Students may be removed from the Co-op Program for any of the following reasons:

1. Failure to achieve a grade of SAT in COOP 1000;
2. Failure to attend all interviews for positions to which the student has applied;
3. Declining more than one job offer during the job search;
4. Reneging on a co-op position that the student has accepted either verbally or in writing;
5. Continuing a job search after accepting a co-op position;
6. Dismissal from a work term by the co-op employer;
7. Leaving a work term without approval from the Co-op Management Team;
8. Receipt of an unsatisfactory work term evaluation;
9. Receiving a grade of UNS on the work term report.

International Students

All international students are required to possess a Co-op Work Permit issued by Immigration, Refugees and Citizenship Canada before they can begin working. The Co-operative Education Office will provide students with a letter of support to accompany their Co-op Work Permit application. Students are advised to discuss the application process and application requirements with the International Student Services Office.

Co-op Fees

All participating Co-op students are required to pay Co-op fees. For full details, please see the Co-op website.

B.A. Honours Geography, Geography with Concentration in Physical Geography, B.Sc. Honours Physical Geography: Co-op Admission and Continuation Requirements

- Maintain full-time status in each study term;
- Be eligible to work in Canada (for off-campus work);
- Have successfully completed COOP 1000 .

In addition to the following:

1. Registered as a full-time student in B.A. Honours Geography, Geography with a Concentration in Physical Geography, Geography with a Concentration in Urban Geography, or B.Sc. Honours Physical Geography;
2. Obtained third-year standing;
3. Successfully completed, by the start-date of the first work term:
 - a. BA students: GEOG 2005, GEOG 2006 and a 0.5 credit from GEOG 3000, GEOG 3010, or GEOG 3030;

- b. B.Sc students: GEOG 2006 and a 0.5 credit from GEOG 3000, GEOG 3010, or GEOG 3030;
4. Obtained an Overall CGPA of at least 9.50 and a Major CGPA of at least 9.50. These CGPAs must be maintained throughout the duration of the degree.

B.A. Honours Geography, Geography with a Concentration in Physical Geography, Geography with a Concentration in Urban Geography, and B.Sc. Honours Physical Geography students must successfully complete three (3) work terms to obtain the Co-op Designation.

Co-op work term course: GEOG 3999

Work/Study Pattern:

Year 1		Year 2		Year 3		Year 4		Year 5	
Term	Pattern	Term	Pattern	Term	Pattern	Term	Pattern	Term	Pattern
Fall	S	Fall	S	Fall	S	Fall	W	Fall	S
Winter	S	Winter	S	Winter	S	Winter	W	Winter	S
Summer		Summer		Summer	W	Summer	W		

Legend

S: Study

W: Work

Admissions Information

Admission Requirements are for the 2026-27 year only, and are based on the Ontario High School System. Holding the minimum admission requirements only establishes eligibility for consideration. The cut-off averages for admission may be considerably higher than the minimum. See also the **General Admission and Procedures** section of this Calendar. An overall average of at least 70% is normally required to be considered for admission. Some programs may also require specific course prerequisites and prerequisite averages and/or supplementary admission portfolios. Higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. The overall average required for admission is determined each year on a program by program basis. Consult admissions.carleton.ca for further details.

Note: Courses listed as *recommended* are not mandatory for admission. Students who do not follow the recommendations will not be disadvantaged in the admission process.

Admissions Information

Admission requirements are based on the Ontario High School System. Prospective students can view the admission requirements through the Admissions website at admissions.carleton.ca. The overall average required for admission is determined each year on a program-by-program basis. Holding the minimum admission requirements only establishes eligibility for consideration; higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. All programs have limited enrolment and admission is not guaranteed. Some programs may also require specific course prerequisites and prerequisite

averages and/or supplementary admission portfolios. Consult admissions.carleton.ca for further details.

Note: If a course is listed as *recommended*, it is not mandatory for admission. Students who do not follow the recommendations will not be disadvantaged in the admission process.

Admission Requirements

Degrees

- Bachelor of Arts (B.A.) (Honours)
- Bachelor of Arts (B.A.)

First Year

For B.A. and B.A. (Honours)

The Ontario Secondary School Diploma (OSSD) or equivalent including a minimum of six 4U or M courses. The six 4U or M courses must include a 4U course in English (or *anglais*). Applicants submitting an English language test to satisfy the requirements of the English Language Proficiency section of this Calendar may use that test to also satisfy the 4U English prerequisite requirement.

Biology

For the major in Biology in the B.A. program, in addition to the 4U English, a 4U course in Chemistry is required. Advanced Functions, and Calculus and Vectors are recommended.

Criminology and Criminal Justice

Access to the CCJ B.A. degree is limited to students already registered in the CCJ B.A. Honours who apply to transfer, and to graduates of the Algonquin College Police Foundations program.

Advanced Standing

Applications for admission beyond first year will be assessed on their merits. Applicants must normally be Eligible to Continue in their year level, in addition to meeting the CGPA thresholds described in Section 3.1.9 of the Academic Regulations of the University. Advanced standing will be granted only for those subjects assessed as being appropriate for the program and the stream selected.

Co-op Option

Direct Admission to the 1st Year of the Co-op Option

Co-op is available for the following Majors in the B.A. (Honours) degree: Anthropology, English, Environmental Studies, European and Russian Studies, French, Geography, Geomatics, History, Human Rights and Social Justice, Law, Political Science, Psychology, Sociology, Women's and Gender Studies.

Applicants must:

1. meet the required overall admission cut-off average and prerequisite course average. These averages may be higher than the stated minimum requirements;
2. be registered as a full-time student in the Bachelor of Arts Honours with one of the majors listed above;
3. be eligible to work in Canada (for off-campus work placements).

Meeting the above requirements only establishes eligibility for admission to the program. The prevailing job market may limit enrolment in the co-op option. Students should also note that hiring priority is given to Canadian citizens for co-op positions in the Public Service Commission.

Note: continuation requirements for students previously admitted to the co-op option and admission requirements for the co-op option after beginning the program are described in the Co-operative Education Regulations section of this Calendar.

Advanced Standing

B.A. and B.A. (Honours) Program

Applications for admission to the second or subsequent years will be assessed on their merits. Advanced standing will be granted only for those courses that are determined to be appropriate.

Admissions Information

Admission Requirements are for the 2026-27 year only, and are based on the Ontario High School System. Holding the minimum admission requirements only establishes eligibility for consideration. The cut-off averages for admission may be considerably higher than the minimum. See also the **General Admission and Procedures** section of this Calendar. An overall average of at least 70% is normally required to be considered for admission. Some programs may also require specific course prerequisites and prerequisite averages and/or supplementary admission portfolios. Higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. The overall average required for admission is determined each year on a program by program basis. Consult admissions.carleton.ca for further details.

Note: Courses listed as *recommended* are not mandatory for admission. Students who do not follow the recommendations will not be disadvantaged in the admission process.

Admissions Information

Admission requirements are based on the Ontario High School System. Prospective students can view the admission requirements through the Admissions website at admissions.carleton.ca. The overall average required for admission is determined each year on a program-by-program basis. Holding the minimum admission requirements only establishes eligibility for consideration; higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. All programs have limited enrolment and admission is not guaranteed. Some programs may also require specific course prerequisites and prerequisite averages and/or supplementary admission portfolios. Consult admissions.carleton.ca for further details.

Note: If a course is listed as recommended, it is not mandatory for admission. Students who do not follow

the recommendations will not be disadvantaged in the admission process.

Degrees

- B.Sc. (Honours)
- B.Sc. (Major)
- B.Sc.

Admission Requirements

B. Sc. Honours

First Year

The Ontario Secondary School Diploma (OSSD) or equivalent including a minimum of six 4U or M courses. For most programs including Biochemistry, Bioinformatics, Biotechnology, Chemistry, Combined Honours in Biology and Physics, Chemistry and Physics, Computational Biochemistry, Food Science, Nanoscience, Neuroscience and Biology, Neuroscience and Mental Health, and Psychology, the six 4U or M courses must include Advanced Functions, and two of Biology, Chemistry, Earth and Space Sciences, or Physics. (Calculus and Vectors is strongly recommended).

Specific Honours Admission Requirements

For the Honours programs in Earth Sciences, Environmental Science, Geomatics, Integrated Science, and Physical Geography, Calculus and Vectors may be substituted for Advanced Functions.

For the Honours programs in Physics and Applied Physics, and for double Honours in Mathematics and Physics, Calculus and Vectors is required in addition to Advanced Functions and one of 4U Physics, Chemistry, Biology, or Earth and Space Sciences. For all programs in Physics, 4U Physics is strongly recommended.

For Honours in Psychology, a 4U course in English is recommended.

For Honours in Environmental Science, a 4U course in Biology and Chemistry is recommended.

Advanced Standing

Applications for admission beyond first year will be assessed on their merits. Applicants must normally be *Eligible to Continue* in their year level, in addition to meeting the CGPA thresholds described in Section 3.1.9 of the Academic Regulations of the University. Advanced standing will be granted only for those subjects deemed appropriate for the program and stream selected.

B.Sc. Major and B.Sc.

First Year

The Ontario Secondary School Diploma (OSSD) or equivalent including a minimum of six 4U or M courses. The six 4U or M courses must include Advanced Functions and two of Calculus and Vectors, Biology, Chemistry, Earth and Space Science, or Physics (Calculus and Vectors is strongly recommended). For the B.Sc. Major in Physics, 4U Physics is strongly recommended.

Advanced Standing

Applications for admission beyond first year will be assessed on their merits. Applicants must normally

be *Eligible to Continue* (EC) in their year level. Advanced standing will be granted only for those subjects deemed appropriate for the program and stream selected.

Co-op Option

Direct Admission to the First Year of the Co-op Option
Applicants must:

1. meet the required overall admission cut-off average and prerequisite course average. These averages may be higher than the stated minimum requirements;
2. be registered as a full-time student in the Bachelor of Science Honours program;
3. be eligible to work in Canada (for off-campus work placements).

Note that meeting the above requirements only establishes eligibility for admission to the program. The prevailing job market may limit enrolment in the co-op option.

Note: continuation requirements for students previously admitted to the co-op option and admission requirements for the co-op option after beginning the program are described in the Co-operative Education Regulations section of this Calendar.

Geography (GEOG) Courses

4000-level courses are normally restricted to students with fourth-year Honours standing. However, students with third-year standing may take 4000-level courses provided they have the necessary prerequisites, a Geography CGPA of 6.50 or better, and permission of the Department.

GEOG 1010 [0.5 credit]

Global Environmental Systems

Introduction to physical and environmental geography. Emphasis on the flow of energy and matter within and between the Earth's atmosphere, hydrosphere, geosphere, and biosphere that shape our physical environment locally and globally. Impacts of climate change and other human activities on environmental systems.

Includes: Experiential Learning Activity

Lectures three hours a week, laboratory two hours a week.

GEOG 1020 [0.5 credit]

People, Places and Environments

Introduction to human geography. Examination of relationships between people, communities, society and the natural environment at local to global scales. Population change, cultural patterns, and historical, economic, political and environmental forces, including climate change, that shape human activity and experiences from place to place.

Includes: Experiential Learning Activity

Also listed as ENST 1020.

Lectures two hours a week and tutorial one hour a week.

GEOG 1023 [0.5 credit]

Introduction to Cities and Urbanization

Geographies of urban experience, development, and change across an urbanizing planet. Historical and contemporary urbanization processes, patterns, and issues in and between cities and regions. The role of urbanization in producing and responding to climate change.

Includes: Experiential Learning Activity

Precludes additional credit for GEOG 2400 (no longer offered).

Lectures two hours per week and tutorials one hour per week.

GEOG 2005 [0.5 credit]

Introduction to Qualitative Research

Introduction to the research process, from generating questions to reporting results. Topics include intensive and extensive research approaches; the use of surveys, interviews and other data collection methods; the analysis of qualitative information; and the ethical dimensions of doing research with people and communities.

Includes: Experiential Learning Activity

Also listed as ENST 2005.

Prerequisite(s): 1.0 credit in GEOG or ENST at the 1000-level and second-year standing, or permission of the Department.

Lectures two hours a week, workshop two hours a week.

GEOG 2006 [0.5 credit]

Introduction to Quantitative Research

Introduction to solving problems using descriptive and inferential statistical methods. Graphical and numerical tools to describe distributions. Probability, sampling and estimates, and hypothesis testing. Fundamentals of spatial statistics and analysis.

Includes: Experiential Learning Activity

Also listed as ENST 2006.

Precludes additional credit for BIT 2000, BIT 2009, BIT 2100 (no longer offered), BIT 2300 (no longer offered), DATA 1517, ECON 2210, NEUR 2002, PSCI 2702, STAT 2507, STAT 2601, STAT 2606.

Lectures two hours a week, laboratory two hours a week.

GEOG 2013 [0.5 credit]

Weather and Water

Introduction to climate, weather and the hydrological cycle. Physical properties of the atmosphere, radiation and energy balances, global circulation, atmospheric moisture and precipitation, weather systems and forecasting, mechanisms of anthropogenic climate change.

Includes: Experiential Learning Activity

Prerequisite(s): GEOG 1010 or EARTH 1002 or ISCI 1001.

Lectures three hours a week, laboratory three hours a week.

GEOG 2014 [0.5 credit]**The Earth's Surface**

Introduction to geomorphology. Weathering, slope and fluvial processes within drainage basins, and glacial and periglacial processes.

Includes: Experiential Learning Activity

Prerequisite(s): GEOG 1010 or EARTH 1002 or ISCI 1001.

Lectures three hours a week, laboratory three hours a week.

GEOG 2020 [0.5 credit]**Ecosystems of Canada**

Introduction to world biomes and in-depth analysis of the characteristics and distribution of Canada's major ecosystems including the boreal forest, temperate forest, tundra, grasslands, wetlands, and aquatic environments; Current issues in ecosystem science and conservation such as climate change, agricultural management, forestry and urban ecology.

Prerequisite(s): GEOG 1010 or EARTH 1002 or EARTH 1004 or ISCI 1001.

Lectures three hours a week.

GEOG 2023 [0.5 credit]**Cities, Inequality and Urban Change**

Geographical perspectives on the uneven power relationships and politics that shape urban lives and urban space. Key topics may include housing and segregation, planning for sustainable cities, urban social movements, urban inequality and changing livelihoods.

Includes: Experiential Learning Activity

Precludes additional credit for GEOG 2400 (no longer offered).

Prerequisite(s): GEOG 1023, or second-year standing, or permission of the department.

Lectures two hours per week and tutorials one hour per week.

GEOG 2200 [0.5 credit]**Global Connections**

Globalization and global environmental change as linked processes. Geographical analysis of economic, cultural, political, and climate change transformations acting at global, national and local scales. Choices and constraints underlying economic, social and environmental sustainability.

Prerequisite(s): second-year standing or permission of the Department.

Lectures three hours a week.

GEOG 2300 [0.5 credit]**Space, Place and Culture**

Introduction to social and cultural geography, including how theories of space, place, landscape, power, and knowledge can be used to understand the geographic dimensions of social and cultural life. Topics include culture and identity, migration and transnationalism, nature, gender, sexuality, race, colonialism, consumption, and work.

Prerequisite(s): second-year standing or permission of the Department.

Lectures two hours a week, discussion one hour a week.

GEOG 2500 [0.5 credit]**Climate Change: Social Science Perspectives**

An introduction to climate change as a political, economic and socio-cultural phenomenon, including the political-economic and world-historical causes of anthropogenic greenhouse gas emissions; variations in impact and vulnerability; climate justice and other political movements; global mitigation and adaptation strategies; and proposals for radical systemic change.

Includes: Experiential Learning Activity

Also listed as ENST 2500.

Prerequisite(s): second-year standing or permission of the Department.

Lectures two hours a week, discussion groups one hour a week.

GEOG 2600 [0.5 credit]**Geography Behind the Headlines**

Exploration of the geographical backgrounds to selected issues of current public interest, through geography's perspective of integrating human and physical environments. Issues selected will be structured from the global through the national/regional to the local, identifying the interdependencies among the scales.

Lecture three hours a week.

GEOG 3000 [0.5 credit]**Honours Field Course**

Field research, with a focus on data collection methods, analysis and presentation of findings. Design and conduct research that links the human and biophysical environment. Topics may change from year to year.

Includes: Experiential Learning Activity

Also listed as ENST 3900.

Precludes additional credit for ENST 2900 (no longer offered).

Prerequisite(s): GEOG 2005/ENST 2005 and GEOG 2006/ENST 2006, third-year Honours standing in Geography, Geomatics or Environmental Studies, or permission of the Department.

Normally consists of a multi-day field excursion in the Ottawa region. A supplementary charge may apply.

Consult the department regarding course details.

GEOG 3001 [0.5 credit]**Doing Qualitative Research**

Theory and methods used in qualitative approaches to research in human geography; hands-on experience and discussion of beliefs and claims underlying scholarly work. Ethical and practical dilemmas confronting researchers. Gathering and interpreting qualitative information; representing knowledge.

Includes: Experiential Learning Activity

Prerequisite(s): GEOG 2005 or ENST 2005.

Lecture and discussion three hours per week.

GEOG 3003 [0.5 credit]**Quantitative Geography**

Quantitative methods used in geographical research: multiple correlation and regression, principal component/factor analysis, spatial statistics, cluster analysis, and a review of other selected techniques. Computer-based analysis.

Includes: Experiential Learning Activity

Prerequisite(s): GEOG 2006 or ENST 2006 or STAT 2507 or permission of the Department.

Lecture two hours a week, laboratory two hours a week.

GEOG 3009 [0.5 credit]**Special Topics in Human Geography**

Selected topics concerning human geography not usually included in regular course offerings. Topic varies from year to year. Students should check with the Department for more information.

Precludes additional credit for GEOG 2505 (no longer offered).

Prerequisite(s): GEOG 1020 or ENST 1020 and third-year standing, or permission of the Department.

Lecture three hours per week.

GEOG 3010 [0.5 credit]**Field Methods in Physical Geography**

Field and laboratory approaches, methodologies and techniques in physical geography. Field projects will be undertaken to collect data for analysis, evaluation and presentation.

Includes: Experiential Learning Activity

Prerequisite(s): GEOG 2006 or ENST 2006 or STAT 2507 and GEOG 2013 or GEOG 2014 or permission of the Department.

Normally consists of a multi-day field camp, including lodging, during Fall or Winter Break, and regular classroom meetings. A supplementary charge will apply.

GEOG 3021 [0.5 credit]**Geographies of Culture and Identity**

Examination of culture, identity and place over time in different contexts; how colonialism, globalization and other processes have shaped societies; geographies of identity, including gender, ethnicity, race and nationality; relationships between cultural groups and their natural surroundings and impacts of climate change.

Includes: Experiential Learning Activity

Prerequisite(s): GEOG 2300 and third-year standing, or permission of the Department.

Lecture three hours a week.

GEOG 3022 [0.5 credit]**Environmental and Natural Resources**

Exploration of complexity, dynamics, uncertainty and equity issues underpinning environmental and resource issues; review and appraisal of selected contemporary methods to assess and manage environmental and natural resources.

Includes: Experiential Learning Activity

Also listed as ENST 3022.

Prerequisite(s): third-year standing in Geography or Environmental Studies or BGIInS Specialization/Stream in Globalization and Environment or permission of the Department.

Lecture three hours a week.

GEOG 3023 [0.5 credit]**Cities in a Global World**

Introduces the study of cities as "systems of cities", the political economy of linkages between urban places located unevenly in space, and "cities as systems". Case studies of socio-cultural, political and economic relations within biophysical and built environments.

Includes: Experiential Learning Activity

Prerequisite(s): GEOG 2023 and third-year standing, or permission of the department.

Lecture and discussion three hours a week.

GEOG 3024 [0.5 credit]**Understanding Globalization**

Geographical analysis of processes of globalization: theoretical frameworks, historical context and contemporary challenges.

Prerequisite(s): GEOG 2200 and third-year standing, or permission of the Department.

Lecture three hours a week.

GEOG 3025 [0.5 credit]**Geographies of Selected Regions**

Geographical analysis of key questions facing a selected region of the world. Attention will focus on selected topics within one or more regions and their related global context.

Prerequisite(s): third-year standing in a B.A. program or BGLnS Specialization/Stream in Globalization and Environment or permission of the Department.
Lecture three hours a week.

GEOG 3026 [0.5 credit]**Topics in the Geography of Canada**

Selected topic concerning the geography of Canada. Topic varies from year to year.

Precludes additional credit for GEOG 2505 [no longer offered].

Prerequisite(s): GEOG 1020 or ENST 1020 and second-year standing, or permission of the Department.
Lecture three hours a week.

GEOG 3030 [0.5 credit]**Regional Field Excursion**

Guided and independent geographic field research, with a focus on data collection methods, and analysis and presentation of findings. Consists of an excursion outside of the Ottawa region. A supplementary charge may apply. Includes: Experiential Learning Activity

Prerequisite(s): third-year Honours standing in Geography or BGLnS Specialization in Globalization and Environment or permission of the Department.
A seven- to ten-day field excursion.

GEOG 3102 [0.5 credit]**Geomorphology**

Geomorphological agents of landscape change at the Earth's surface, emphasizing the role of water, ice and wind in erosion and deposition; use of geomorphic indicators in studies of environmental and climate change. A supplementary charge may apply.

Includes: Experiential Learning Activity
Prerequisite(s): GEOG 2014 and third-year standing, or permission of the Department.
Lectures two hours a week, laboratory two hours a week, one field excursion.

GEOG 3103 [0.5 credit]**Watershed Hydrology**

Principles of watershed hydrology and climate change impacts on renewable freshwater resources, emphasizing the physical mechanisms of runoff generation, groundwater flow, soil water movement, evapotranspiration and snowmelt.

Includes: Experiential Learning Activity

Prerequisite(s): GEOG 2013 or permission of the Department.

Lectures three hours a week, laboratory two hours a week.

GEOG 3104 [0.5 credit]**Principles of Biogeography**

Contemporary and past controls on distribution of plants and animals at global, regional and local scales; significance of these distributions.

Includes: Experiential Learning Activity

Also listed as BIOL 3608.

Prerequisite(s): GEOG 1010 or BIOL 2600, or permission of the Department.

Lectures, laboratory, and fieldwork five hours a week.

GEOG 3105 [0.5 credit]**Climate and Atmospheric Change**

The global climate system, with emphasis on global change variability over the historical and modern periods; the changing composition of the atmosphere and its impact on climate; analysis and interpretation of climatic and atmospheric data; modeling of climate systems.

Includes: Experiential Learning Activity

Prerequisite(s): GEOG 2013 or permission of the Department.

Lecture two hours a week, laboratory two hours a week.

GEOG 3106 [0.5 credit]**Aquatic Science and Management**

Fundamentals of aquatic science. The physical, chemical, and biotic aspects of lake, river, and estuary ecosystems in different geographic contexts. The impacts of climate and environmental changes on aquatic ecosystems and management and conservation actions to conserve and restore these ecosystems.

Includes: Experiential Learning Activity

Also listed as ENSC 3106.

Prerequisite(s): third-year standing and a second-year science or engineering course.

Lectures three hours per week.

GEOG 3108 [0.5 credit]**Soil Properties**

The physical, chemical and biological properties of soils and how these impact soil-water relationships, soil quality, soil fertility, soil biology and soil classification, among other topics. Examines the role of soils in food production and climate change.

Includes: Experiential Learning Activity

Prerequisite(s): GEOG 2013 or GEOG 2014 or permission of the Department.

Lectures and laboratory five hours a week.

GEOG 3206 [0.5 credit]**Health, Environment, and Society**

Explores the nexus between geography and human health, including climate change impacts on disease, mental and physical health, and inequity; relationships between colonization, modernization, identity, ideologies, and the environment; population health and health behaviour; social determinants of global health inequality and possibilities for change.

Prerequisite(s): third-year standing.

Lectures three hours a week.

GEOG 3209 [0.5 credit]**Sustainability and Environment in the South**

Analysis of the relationships between people and environment in selected regions in the South (Africa, Asia, Latin America). Emphasis on sustainable livelihoods and local action in relation to climate change and broader socioeconomic and political processes. Regions selected vary from year to year.

Prerequisite(s): third-year standing and ENST 2000 or ENST 2001 or GEOG 2200 or GEOG 2300 or permission of the Department.

Lecture and discussion three hours a week.

GEOG 3404 [0.5 credit]**Geographies of Economic Development**

Geographical approaches to economic development and difference at local, regional, and global scales. Critical historical, cultural, social, political, economic, and environmental perspectives on 'development', including theories of the state, colonial power, development institutions, and climate change. Spatial dynamics and environmental impacts of economic activity.

Prerequisite(s): GEOG 2200 or permission of the Department.

Lectures three hours a week.

GEOG 3501 [0.5 credit]**Geographies of the Canadian North**

Key issues in contemporary northern Canada, including land and jurisdiction, wildlife, resource extraction, economic development, culture, geopolitics, health, and climate change. Historical geography and physical characteristics of the region.

Prerequisite(s): third-year standing or permission of the Department.

Lectures three hours a week.

GEOG 3700 [0.5 credit]**Population Geography**

The distributional aspects of population attributes; areal patterns of population characteristics and their spatial variations associated with differences in the nature of places; migratory movements within the framework of spatial models of interactions between locations.

Prerequisite(s): GEOG 2200 or GEOG 2300, or permission of the Department.

Lectures three hours a week.

GEOG 3999 [0.0 credit]**Co-operative Work Term**

Includes: Experiential Learning Activity

GEOG 4000 [0.5 credit]**Field Studies**

Field observation and methodology in a selected region; individual or group basis.

Includes: Experiential Learning Activity

Also listed as ENST 4400.

Prerequisite(s): third-year Honours standing and permission of the Department.

Hours to be arranged.

GEOG 4004 [0.5 credit]**Environmental Impact Assessment**

Principles, scope and purpose of environmental impact assessment, from conceptual and methodological points of view; range of environmental issues, with emphasis on Canadian case studies.

Includes: Experiential Learning Activity

Also listed as ENST 4004.

Prerequisite(s): GEOG 3022 or ENST 3022, and fourth-year Honours standing in Geography or Environmental Studies or Environmental Science, or permission of the Department.

Lectures and seminars three hours a week.

GEOG 4005 [0.5 credit]**Directed Studies in Geography**

Students pursue their interest in a selected theme in geography on a tutorial basis with a member of the Department.

Prerequisite(s): permission of the Department.

GEOG 4007 [0.5 credit]**Special Topics in Geography and Environmental Studies**

Selected topics in geography and/or environmental studies.

Also listed as ENST 4007.

Precludes additional credit for GEOG 4006.

Prerequisite(s): fourth-year Honours standing in the Department or permission of the Department.

Seminar three hours per week.

GEOG 4013 [0.5 credit]**Cold Region Hydrology**

An examination of cold region hydrologic processes via experimental and observational studies emphasizing arctic, sub-arctic and northern boreal landscapes experiencing rapid climate change.

Prerequisite(s): GEOG 3103.

Lecture three hours a week.

GEOG 4017 [0.5 credit]**Global Biogeochemical Cycles**

Processes that control the fluxes and reservoirs of biologically active chemical constituents on land, in the atmosphere, and in the oceans. Climate change impacts on global carbon and nitrogen cycles and their feedbacks on the climate system.

Prerequisite(s): GEOG 3108 or permission of the Department.

Lectures three hours a week.

GEOG 4021 [0.5 credit]**Seminar in Culture, Identity and Place**

Selected topic or field of inquiry concerning the geographic dimensions of culture, identity and place.

Prerequisite(s): GEOG 3021 and fourth-year Honours standing in Geography or permission of the Department.

Seminar three hours a week.

GEOG 4022 [0.5 credit]**Seminar in People, Resources and Environmental Change**

A selected topic or field of inquiry concerning natural resource use and environmental change.

Also listed as ENST 4022.

Prerequisite(s): GEOG 3022 or ENST 3022 and fourth-year Honours standing in Geography or Environmental Studies or BGIInS Specialization in Globalization and Environment or permission of the Department.

Seminar three hours a week.

GEOG 4023 [0.5 credit]**Seminar in Special Topics on the City**

A selected topic or field of inquiry concerning urban geography.

Prerequisite(s): GEOG 3023 and fourth-year Honours standing in Geography or Environmental Studies or BGIInS Specialization in Globalization and Environment or permission of the Department.

Seminar three hours per week.

GEOG 4024 [0.5 credit]**Seminar in Globalization**

A selected issue or topic related to globalization.

Prerequisite(s): GEOG 3024 and fourth-year Honours standing in Geography or BGIInS Specialization in Globalization and Environment or permission of the Department.

Seminar three hours week.

GEOG 4040 [0.5 credit]**Geographic Thought**

Major intellectual issues and debates in the development of contemporary human geography, including history of geographic thought, geographic responses to social and political movements and debates, and geographic engagement with contemporary critical theory.

Prerequisite(s): fourth-year Honours standing in Geography or permission of the Department.

Seminar three hours per week.

GEOG 4050 [0.5 credit]**Environmental and Geographic Education**

Selected theoretical and applied issues concerning environmental and geographic education.

Also listed as ENST 4050.

Prerequisite(s): third-year Honours standing in Geography or Environmental Studies, or permission of the Department.

Seminar three hours per week.

GEOG 4101 [0.5 credit]**Two Million Years of Environmental Change**

Multidisciplinary scientific study of the changes in the physical environment of the Earth during the last two million years and methods of studying recent Earth history, with focus on current research.

Includes: Experiential Learning Activity

Prerequisite(s): third year standing in a B.Sc. program, or a third year Science Geography Elective or a third year EARTH course, or permission of the Department. Note: GEOG 3105 is recommended.

Lectures three hours a week.

GEOG 4103 [0.5 credit]**Water Resources Engineering**

A quantitative analysis of natural water systems and the development of these systems as a resource.

Components of the hydrologic cycle. Quantitative analysis of stream flow. Probability concepts in water resources. Reservoir design and operation. Availability of groundwater. Storm water management.

Also listed as ENVE 3003.

Prerequisite(s): permission of the Department.

Recommended background: MAAE 2300.

Lectures three hours a week, problem analysis one hour a week.

GEOG 4104 [0.5 credit]**Microclimatology**

The formation of microclimates near the Earth's surface.

Discussion and demonstration of techniques used to measure and monitor microclimates. Microclimate impacts on forest, crop and animal production, hydrology, urban heat islands, and the impacts of climate change.

Prerequisite(s): GEOG 2013 or permission of the Department.

Lectures three hours a week.

GEOG 4108 [0.5 credit]**Permafrost**

Understanding permafrost processes and phenomena and how they respond to climate change. Topics include the distribution, development, and degradation of permafrost; interactions between atmosphere, snow, and ground; the thermal and hydrologic regime of permafrost terrain; landforms in permafrost regions; geotechnical consideration in northern construction.

Prerequisite(s): GEOG 3108 or permission of the Department.

Lectures three hours a week.

GEOG 4304 [0.5 credit]**Transportation Engineering and Planning**

Transportation and the socio-economic environment; modal and intermodal systems and components; vehicle motion; human factors, system and facility design; traffic flow; capacity analysis; planning methodology; environmental impacts; evaluation methods.

Precludes additional credit for CIVE 3304.

Prerequisite(s): third-year standing, or permission of the Department.

Lectures three hours a week, problem analysis three hours alternate weeks.

GEOG 4323 [0.5 credit]**Urban and Regional Planning**

History, theories, and practice of urban planning, as well as the policies, plans, and programs developed and implemented in diverse communities. Course topics may include the integration of community development and social planning, urban design, transportation and infrastructure, and environmental management.

Includes: Experiential Learning Activity

Prerequisite(s): GEOG 3023 and fourth-year standing in Geography or Environmental Studies, or permission of the department.

Lectures three hours per week.

GEOG 4406 [0.5 credit]**Practicum I**

Students apply their knowledge and research skills and gain experience through field placements in government, the private sector, non-government organisations and with community organisations in the environmental field.

Includes: Experiential Learning Activity

Also listed as GEOM 4406.

Prerequisite(s): fourth-year Honours standing in Geography or Geomatics and permission of the Department.

Field placement one day a week.

GEOG 4408 [0.5 credit]**Practicum II**

Students apply their knowledge and research skills and gain experience through field placements in government, the private sector, non-government organisations and with community organisations in the environmental field.

Includes: Experiential Learning Activity

Also listed as GEOM 4408.

Prerequisite(s): fourth-year Honours standing in Geography or Geomatics and permission of the Department.

Field placement of one day a week.

GEOG 4450 [0.5 credit]**Community-Engaged Research**

Working in partnership with local organizations, students apply their geographical knowledge to conduct community-engaged research. Student projects will generate outputs for community partners. Research topics vary year to year.

Includes: Experiential Learning Activity

Also listed as ENST 4450.

Prerequisite(s): fourth-year standing, or permission of the department.

Lectures, discussion and project work three hours a week.

GEOG 4906 [1.0 credit]**Honours Research Project**

A research project based on a modeling, laboratory or field problem. The project is supervised by a member of the department and a written thesis and poster must be submitted.

Includes: Experiential Learning Activity

Precludes additional credit for GEOG 4904/GEOM 4904 (no longer offered), GEOM 4906, GEOG 4909, GEOM 4909, ENST 4906, and ENST 4907.

Prerequisite(s): fourth-year Honours standing in B.Sc. Geography, and an approved research topic and adviser.

Hours to be arranged with faculty adviser.

GEOG 4909 [1.0 credit]**Honours Research Thesis**

Independent design and implementation of a research project leading to the submission of a research thesis.

Students work with an individual faculty adviser. The subject for research is decided upon in consultation with the supervisor.

Includes: Experiential Learning Activity

Precludes additional credit for GEOG 4904/GEOM 4904 (no longer offered), GEOG 4906, GEOM 4906, GEOM 4909, ENST 4906, and ENST 4907.

Prerequisite(s): fourth-year Honours standing in B.A. Geography or B.Globalization and International Studies, a minimum CGPA of 9.00 in the major or permission of the Department, and an approved research topic and adviser.

Hours to be arranged with faculty adviser.